

SURAJ S WARRIER

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SUMMARY

Software Development Engineer with 1+ years of experience designing scalable backend microservices, distributed data pipelines, and graph-based intelligence systems in C++ and Python. Reduced production latency by up to 91% and cut query response times by 40% through system-level optimization, graph architectures, and cloud-native infrastructure (Docker, Kubernetes, Kafka).

EDUCATION

Indian Institute of Technology Patna

2021 – 2025

B.Tech Computer Science · CPI: 8.98 / 10

EXPERIENCE

R&D Engineer

Jul 2025 – Present

Samsung Research Institute Noida

Noida, India

- Architected a distributed graph-based policy orchestration subsystem, converting enterprise rulesets into dependency graphs and applying vector retrieval to resolve execution conflicts and eliminate device loops.
- Engineered a code-intelligence graph engine using Abstract Syntax Tree (AST) extraction, Neo4j, and ChromaDB across 10K+ source files, cutting developer query lookup time by 40% for 50+ engineers.
- Reduced on-device inference latency by 91% (from 7–8 minutes to 40 seconds) by building custom prompt-caching, weight quantization, and native runtime profiling pipelines with llama.cpp.

Full Stack Developer Intern

May – Jul 2024

ThinkPalm Technologies Kochi

Kochi, India

- Delivered a production financial forecasting platform on a .NET Core and React stack, feeding time-series data through a custom LSTM model to improve forecast accuracy by 18%.
- Redesigned backend microservices powering live market data feeds, restructuring endpoint communication to cut REST API latency by 35%.

PROJECTS

LogHelmsman – Kubernetes Anomaly Detection

Apr 2026 – May 2026

- Built a real-time, event-driven infrastructure monitoring pipeline using Fluent Bit and Apache Kafka, routing metrics to a FastAPI service that flags cluster anomalies within 2–5 minutes of failure.
- Trained an Isolation Forest model on per-pod behavioral metrics and paired it with schema-driven correlation logic to detect crash loops, resource exhaustion, and cascading faults.
- Scaled event throughput with parallel Kafka consumer groups and FastAPI multiprocessing, enabling concurrent handling across multiple high-load cluster nodes.

Video Chat Web Application · React, Node.js, Socket.io, Agora, MERN Stack

Sept 2023 – Dec 2023

- Launched a full-stack live-collaboration platform supporting concurrent media pipelines, achieving sub-100ms message delivery and a 200ms connection setup.
- Developed connection event listeners to clean up idle sessions and drop dead channels, preventing memory leaks during traffic spikes.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, JavaScript

Frameworks & Web: React, Node.js, Express.js, FastAPI, .NET Core, Socket.io

Tools & Infrastructure: Docker, Kubernetes, Apache Kafka, Neo4j, ChromaDB, Git, Perforce

Core Concepts: Data Structures & Algorithms, System Design, Distributed Systems, REST API Design, Microservices Architecture

Coursework: Operating Systems, Computer Networks, OOP, DBMS

ACHIEVEMENTS & LEADERSHIP

- Codeforces Expert** (Max Rating: 1607).
- LeetCode Knight** (Max Rating: 1984).
- CodeChef 4★** (Max Rating: 1836).
- JEE Advanced 2021:** All India Rank **2598** out of 160,000+ competing candidates.
- Leadership:** Core Committee Member, Aria Music Club; Volunteer Lead for Anwesha — managed operations and core logistics for over 5,000+ festival attendees.